THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** rotation	Spinning.
2	**(d)** Purnima	Full moon day.
3	**(b)** Earth rotates	Apparent motion.
4	**(d)** 24 hours	A day.
5	**(c)** Amavasya	New moon.
6	**(a)** Pole Star	Near the axis.
7	**(b)** one year	Around the Sun.
8	**(d)** 29.5 days	Lunar month.
9	**(c)** constellation	Pattern.
10	**(d)** 366 days	Extra day.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rotation of the Earth (2).	2
12	Time between two successive new moons, about 29.5 days (2).	2
13	Spinning of the Earth on its axis (2).	2
14	A year of 366 days (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Day and night (1); Apparent motion of the Sun (1); Apparent motion of stars (1).	1+1+1
16	Natural satellite (1); Reflects sunlight (1); Shows phases (1).	1+1+1
17	Sundial uses Sun (1); Watch uses a mechanism (1); Sundial does not work at night (1).	1+1+1
	OR: Year (1); Seasons (1); Change in the night sky (1).	1+1+1
18	Fixed in the northern sky (1); Shows north (1); Lies near the Earth's axis (1).	1+1+1
	OR: Stars (1); Moon (1); Planets (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rotation: 24 hours (1); Revolution: 365 days (1); Day and night (1); Seasons (1); Leap year (1).	1+1+1+1+1
	OR: Morning (1); Noon (1); Evening (1); Direction (1); Length (1).	1+1+1+1+1
20	New moon (1); Crescent (1); Half moon (1); Full moon (1); Waning (1).	1+1+1+1+1
	OR: 21 June (1); 22 December (1); 21 March (1); 23 September (1); Meaning (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 27 days	Orbit.
2	**(c)** Big Dipper or Ursa Major	Seven stars.
3	**(a)** east	East.
4	**(b)** 23.5 degrees	Tilt.
5	**(d)** Jantar Mantar	Built by Jai Singh.
6	**(b)** west	West.
7	**(a)** tilt of Earth's axis	Tilt and revolution.
8	**(b)** star	Our star.
9	**(d)** north direction	North.
10	**(c)** 29 days	Extra day.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Different shapes of the Moon seen from Earth (2).	2
12	Movement of the Earth around the Sun (2).	2
13	A device that shows time by the shadow of a stick (2).	2
14	Nearly circular (elliptical) (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Long in the morning (1); Short at noon (1); Long in the evening (1).	1+1+1
16	Sundial (1); Noon shadow (1); Day and night (1).	1+1+1
17	Located in Jaipur (1); Built by Sawai Jai Singh II (1); Has large stone instruments (1).	1+1+1
	OR: New moon (1); First quarter (1); Full moon (1).	1+1+1
18	Summer (1); Monsoon (1); Winter (1).	1+1+1
	OR: Sun is a star, Moon is a satellite (1); Sun gives light, Moon reflects (1); Sun is much bigger (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Vertical stick (1); Sunny place (1); Mark shadows hourly (1); Join marks (1); Read time (1).	1+1+1+1+1
	OR: In Jaipur (1); Built by Jai Singh (1); Samrat Yantra (1); Measures time (1); Tracks the Sun and stars (1).	1+1+1+1+1
20	Find the Big Dipper (1); Pointer stars (1); Extend the line (1); Locate the Pole Star (1); It shows north (1).	1+1+1+1+1
	OR: Natural satellite (1); Reflects light (1); Orbits the Earth (1); Phases (1); Same face to Earth (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** equinox	Equal length.
2	**(b)** shadow	Shadow of a gnomon.
3	**(a)** Sun	Sun.
4	**(a)** Moon's phases	Moon.
5	**(c)** shortest	Sun is high.
6	**(a)** Moon	Moon.
7	**(c)** west to east	Spinning direction.
8	**(d)** Sun	Not luminous.
9	**(d)** June	Summer solstice.
10	**(a)** an orbit	Elliptical path.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gregorian calendar (1); Hindu calendar (1).	1+1
12	A star in the north that appears fixed (2).	2
13	21 March (1); 23 September (1).	1+1
14	Orion (1); Ursa Major (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Earth takes 365.25 days (1); Extra quarter days add up (1); Adds one day in four years (1).	1+1+1
16	Solstice: longest or shortest day (1); Equinox: equal day and night (1); Dates differ (1).	1+1+1
17	Reflects sunlight (1); Not self-luminous (1); Appears bright at night (1).	1+1+1
	OR: 29.5 days vs 28 to 31 (1); Moon phase based vs fixed (1); Lunar start at new/full moon (1).	1+1+1
18	April (1); June (1); September (1).	1+1+1
	OR: Festivals (1); Exams (1); Holidays (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Solar (1); Lunar (1); Lunisolar (1); Examples (1); Leap year (1).	1+1+1+1+1
	OR: Earth rotates (1); Sun appears to rise and set (1); Stars appear to move (1); Pole Star fixed (1); Star trails (1).	1+1+1+1+1
20	Axis tilted (1); Revolution (1); Direct rays in summer (1); Slanting rays in winter (1); Day length (1).	1+1+1+1+1
	OR: Earth takes 365.25 days (1); Add one day every four years (1); February has 29 days (1); Example 2024 (1); Century rules exist (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.